

NIRAV SURABHI

Software + mechanical engineering, from Civic aerodynamics to bias-reducing news apps.

Brambleton, VA
niravsurabhi@gmail.com
niravsurabhi.com
linkedin.com/in/niravss
github.com/21J3phy

EDUCATION

- Purdue University

FALL 2026 · INCOMING

First-Year Engineering · Intended Mechanical Engineering
- Independence High School

GRAD. JUN 2026

Loudoun County, VA · Class of 2026

Coursework: AP Computer Science A, AP Physics C, Multivariable Calculus.
- Central Virginia Governor's School

JUN-JUL 2024

Science & Technology · Mathematics scholar

RESEARCH & PUBLICATION

- Passive Vortex Propulsion

2024-2025 · 1ST AUTHOR

IEEE Xplore · Document 11567580

Designed and built a wind-tunnel test rig, soldered and calibrated an Arduino load-cell system, ran CFD, and analyzed 20 trials. Measured **3.1 N mean net thrust ($p < 0.001$)**, providing the first experimental evidence of passive vortex propulsion in air.

Arduino · C++ · load cells · Fusion 360 · CFD · Python · statistical analysis

ENGINEERING PROJECTS

- Motor-Based Resistance System

2025 · RSEF · 1ST AUTHOR

Designed and built a portable, battery-powered strength trainer through three prototypes. A 188:1 geared motor produced a repeatable, PWM-controlled force curve (**$R^2 = 0.993$**) at a \$214.95 prototype cost.

Fusion 360 · FEA · Arduino · PWM · motor control · prototyping
- FTC Robotics — Absolute Zero (12096)

2022-2026 · CAPTAIN

Four years as captain & lead programmer; CADded and built every part in Fusion 360. Wrote autonomous & tele-op in Java — PID loops, sensor fusion, odometry. Reached the Chesapeake championship.

SKILLS

PROGRAMMING	Python, Java, C++, JavaScript, TypeScript, Git
ENGINEERING	Fusion 360, CFD, FEA, Arduino, load cells, soldering
CONTROLS	PID, sensor fusion, odometry, motor control, data analysis
AI / ML	PyTorch, computer vision, LLMs, agent architecture

HONORS & MUSIC

- AIAA Aerospace Award (RSEF special honor → NASA Goddard invitation)
- Captain, Science Olympiad
- Presidential Volunteer Service Award (Silver)
- Central Virginia Governor's School for Science & Technology — Mathematics, 2024
- 1st place, Aguado guitar competition · All-Virginia Guitar Ensemble

SOFTWARE / ML — SELECTED

- Khali

2025-NOW · CO-FOUNDER/LEAD ENGINEER

khali.info

TikTok-style swipe-through news startup built to make short-form scrolling useful: stories combine opposing sources, present one claim at a time, and include an OpenAI "ask this story" assistant. Native SwiftUI iOS + Expo/RN app on a Node + Python backend.

Swift/SwiftUI · React Native · Expo · Node.js · Python · OpenAI · Supabase
- Articulator

2026-NOW · CREATOR · INDYHAX

articulator-3000.vercel.app

Turns a rough idea — spoken, typed, air-drawn over webcam (MediaPipe, client-side), or uploaded — into a structured model, interviews you until it understands, runs a web-research reality check, and emits briefs and a concept image. Hackathon project taken to production.

Next.js · TypeScript · OpenAI Responses API · MediaPipe · Supabase · Zod
- Oh

2026-NOW · CREATOR

github.com/21J3phy/Oh

Built shared memory for AI coding agents: hook-driven session capture, local secret scrubbing, vector retrieval, and an MCP interface that preserves the reasoning behind code changes.

TypeScript · Node.js · MCP · OpenAI · Supabase · pgvector
- Chad vs. Chud

2026 · CO-FOUNDER/LEAD ENGINEER

chadvschud.com · sunset

Built a real-time webcam client, Socket.IO/WebRTC matchmaking, and a FastAPI inference service using a fine-tuned ResNet-18 + ArcFace ensemble. **10,000+ users; 100+ concurrent.**

Next.js · TypeScript · FastAPI · WebRTC · PyTorch · Fly.io
- EXPERIENCE
- CAD Intern

JUN 2025-MAY 2026

I3 Simulations

3D models and haptic-feedback prototypes for medical-training simulators.
- Bike Mechanic

JUN 2021-AUG 2025

The Bike Lane · intern → part-time

Built bikes from the frame up; ran final safety checks.
- AI Engineer & SWE Intern

FEB 2021-2022

Ember Learning (formerly DeAP Learning Labs)

Wrote the LLM prompts behind AP-tutor chatbots; built automated AI grading for AP DBQs/FRQs against the official rubric; indexed lecture transcripts in a vector DB for timestamp search; led the company website. **Team landed a Microsoft & OpenAI grant; \$1M+ valuation.**